

Analytic spaces

Fix a complete non-archimedean field $\mathbf{k}$.

1 Definition of analytic spaces

Definition 1. Let X be a topological space. A *net* on X is a collection $\mathcal{N}$ of subsets of X such that

- (a) for every $x \in X$, there exist $U_1, \dots, U_n \in \mathcal{N}$ such that $x \in \bigcap_i U_i$ and $\bigcup_i U_i$ is an open neighborhood of x in X ;
- (b) for every $U, V \in \mathcal{N}$, the collection $\mathcal{N}|_{U \cap V} := \{W \in \mathcal{N} : W \subseteq U \cap V\} \subset \mathcal{P}(U \cap V)$ satisfies the first condition for the topological space $U \cap V$.

Yang: For example, if X is a $\mathbf{k}$ -affinoid space, then the collection of all affinoid domains in X forms a net on X .

Definition 2. Let X be a locally Hausdorff topological space. A $\mathbf{k}$ -*affinoid atlas* on X is a net $\mathcal{N}$ on X together with

- (a) for every $U \in \mathcal{N}$, a structure of a $\mathbf{k}$ -affinoid space on U , i.e. a $\mathbf{k}$ -affinoid algebra A_U and an isomorphism $U \cong \mathcal{M}(A_U)$;
- (b) for every $U, V \in \mathcal{N}$ with $V \subseteq U$, a morphism $\text{res}_V^U : A_U \rightarrow A_V$ of $\mathbf{k}$ -affinoid algebras such that the induced morphism $\mathcal{M}(A_V) \rightarrow \mathcal{M}(A_U)$ making V a $\mathbf{k}$ -affinoid subdomain in U .

A $\mathbf{k}$ -*analytic space* or *Berkovich $\mathbf{k}$ -analytic space* is a locally Hausdorff topological space X together with a $\mathbf{k}$ -affinoid atlas on X .

Remark 3. There are two different versions of the definition of $\mathbf{k}$ -analytic spaces in the literature: one given by Berkovich in

Definition 4. Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be $\mathbf{k}$ -analytic spaces. A *morphism* $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ of $\mathbf{k}$ -analytic spaces is a morphism of locally ringed spaces. The category of $\mathbf{k}$ -analytic spaces is the category whose objects are $\mathbf{k}$ -analytic spaces and whose morphisms are morphisms of $\mathbf{k}$ -analytic spaces. Yang: To be revised.

Theorem 5. Let $(X, \mathcal{O}_X)$, $(Y, \mathcal{O}_Y)$, and $(Z, \mathcal{O}_Z)$ be $\mathbf{k}$ -analytic spaces, and let $f : (X, \mathcal{O}_X) \rightarrow (Z, \mathcal{O}_Z)$ and $g : (Y, \mathcal{O}_Y) \rightarrow (Z, \mathcal{O}_Z)$ be morphisms of $\mathbf{k}$ -analytic spaces. Then the fibre product $(X, \mathcal{O}_X) \times_{(Z, \mathcal{O}_Z)} (Y, \mathcal{O}_Y)$ exists in the category of $\mathbf{k}$ -analytic spaces. Yang: To be revised.

Definition 6. Let $(X, \mathcal{O}_X)$ be a $\mathbf{k}$ -analytic space. We say that $(X, \mathcal{O}_X)$ is *good* if every point $x \in X$ has an open neighborhood U such that $(U, \mathcal{O}_X|_U)$ is a $\mathbf{k}$ -affinoid space.

2 The first properties

Definition 7. Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of $\mathbf{k}$ -analytic spaces. We say that f is *finite* if for every affinoid domain $V \subseteq Y$, the preimage $f^{-1}(V)$ is an affinoid domain in X and the induced morphism $\mathcal{O}_Y(V) \rightarrow \mathcal{O}_X(f^{-1}(V))$ is a finite morphism of $\mathbf{k}$ -affinoid algebras. **Yang:**

Theorem 8. Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of $\mathbf{k}$ -analytic spaces. Then there exists a factorization $f = g \circ h$ where $h : (X, \mathcal{O}_X) \rightarrow (Z, \mathcal{O}_Z)$ is a finite morphism and $g : (Z, \mathcal{O}_Z) \rightarrow (Y, \mathcal{O}_Y)$ is a morphism such that for every affinoid domain $W \subseteq Y$, the preimage $g^{-1}(W)$ is an affinoid domain in Z and the induced morphism $\mathcal{O}_Y(W) \rightarrow \mathcal{O}_Z(g^{-1}(W))$ is a flat morphism of $\mathbf{k}$ -affinoid algebras. **Yang:**

Definition 9. Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of $\mathbf{k}$ -analytic spaces. We say that f is *separated* if the diagonal morphism $\Delta_f : (X, \mathcal{O}_X) \rightarrow (X, \mathcal{O}_X) \times_{(Y, \mathcal{O}_Y)} (X, \mathcal{O}_X)$ is a closed immersion. **Yang:**

Definition 10. Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of $\mathbf{k}$ -analytic spaces. We say that f is *proper* if it is separated, compact, and without boundary. **Yang:**

Theorem 11. Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a proper morphism of $\mathbf{k}$ -analytic spaces. Then for every coherent $\mathcal{O}_X$ -module $\mathcal{F}$, the direct image $f_*(\mathcal{F})$ is a coherent $\mathcal{O}_Y$ -module. **Yang:**

Definition 12. Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of $\mathbf{k}$ -analytic spaces. We say that f is *smooth* (resp. *étale*) if for every $x \in X$, there exist open neighborhoods U of x in X and V of $f(x)$ in Y such that $f(U) \subseteq V$ and a commutative diagram **Yang:**

Appendix